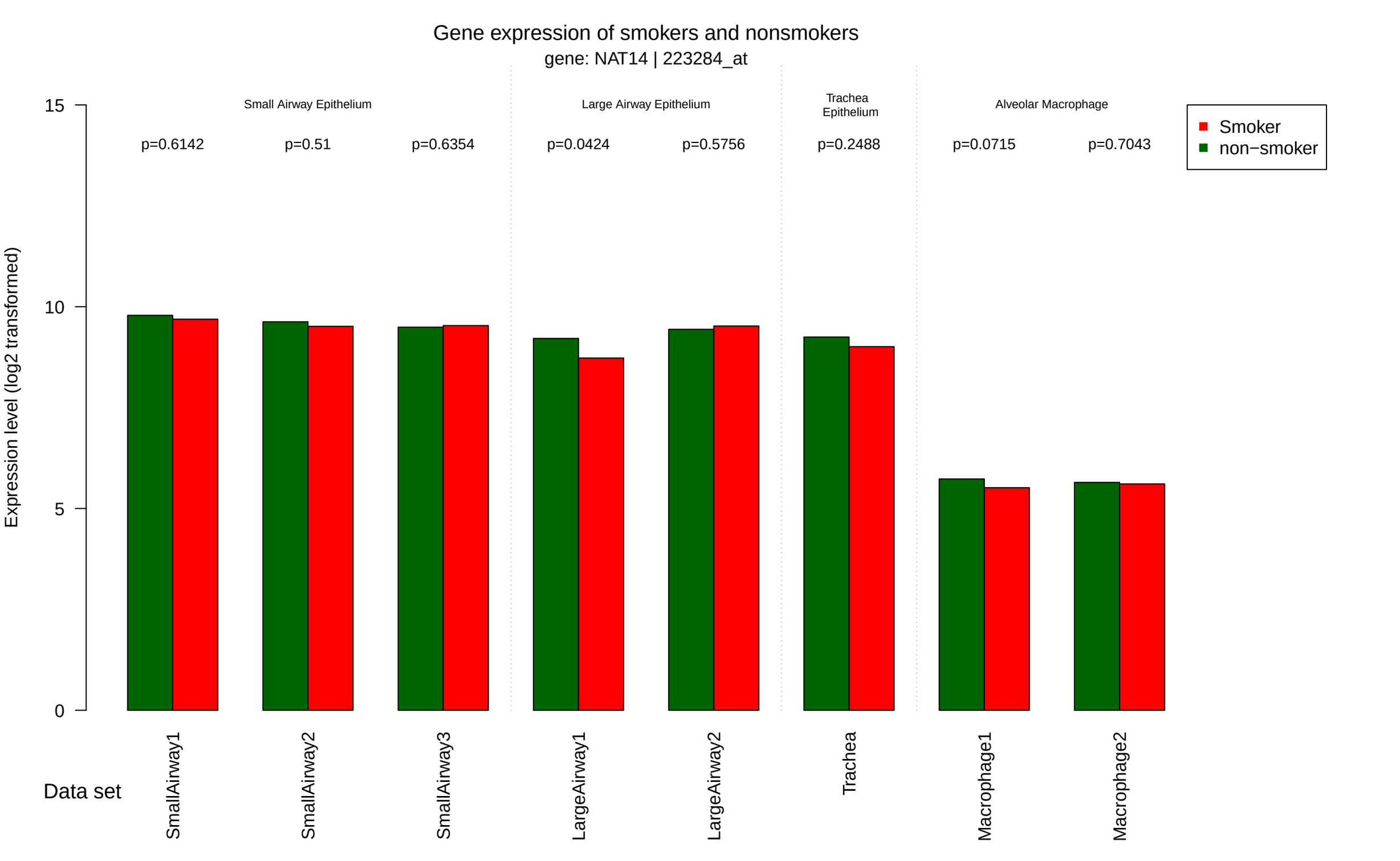
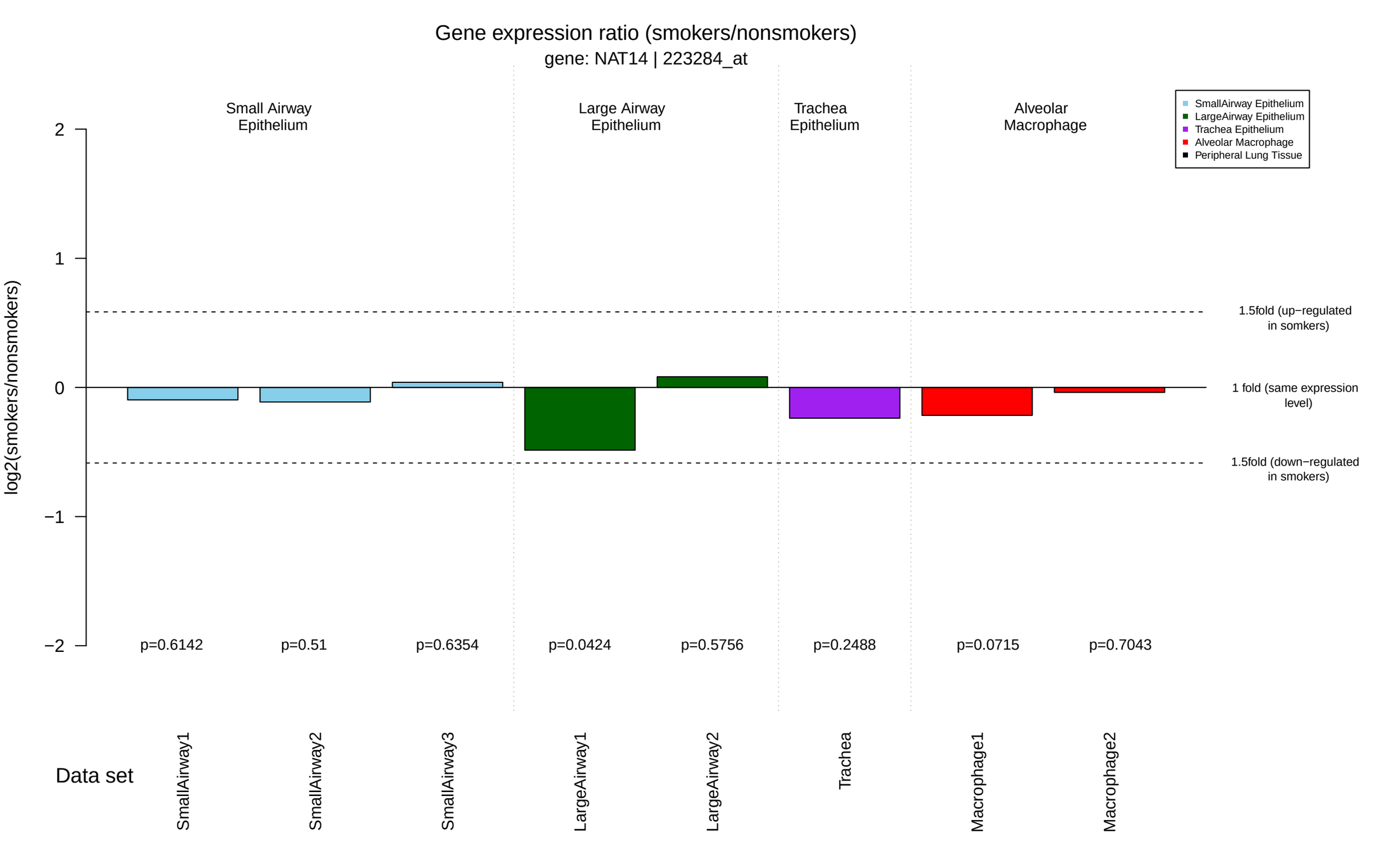
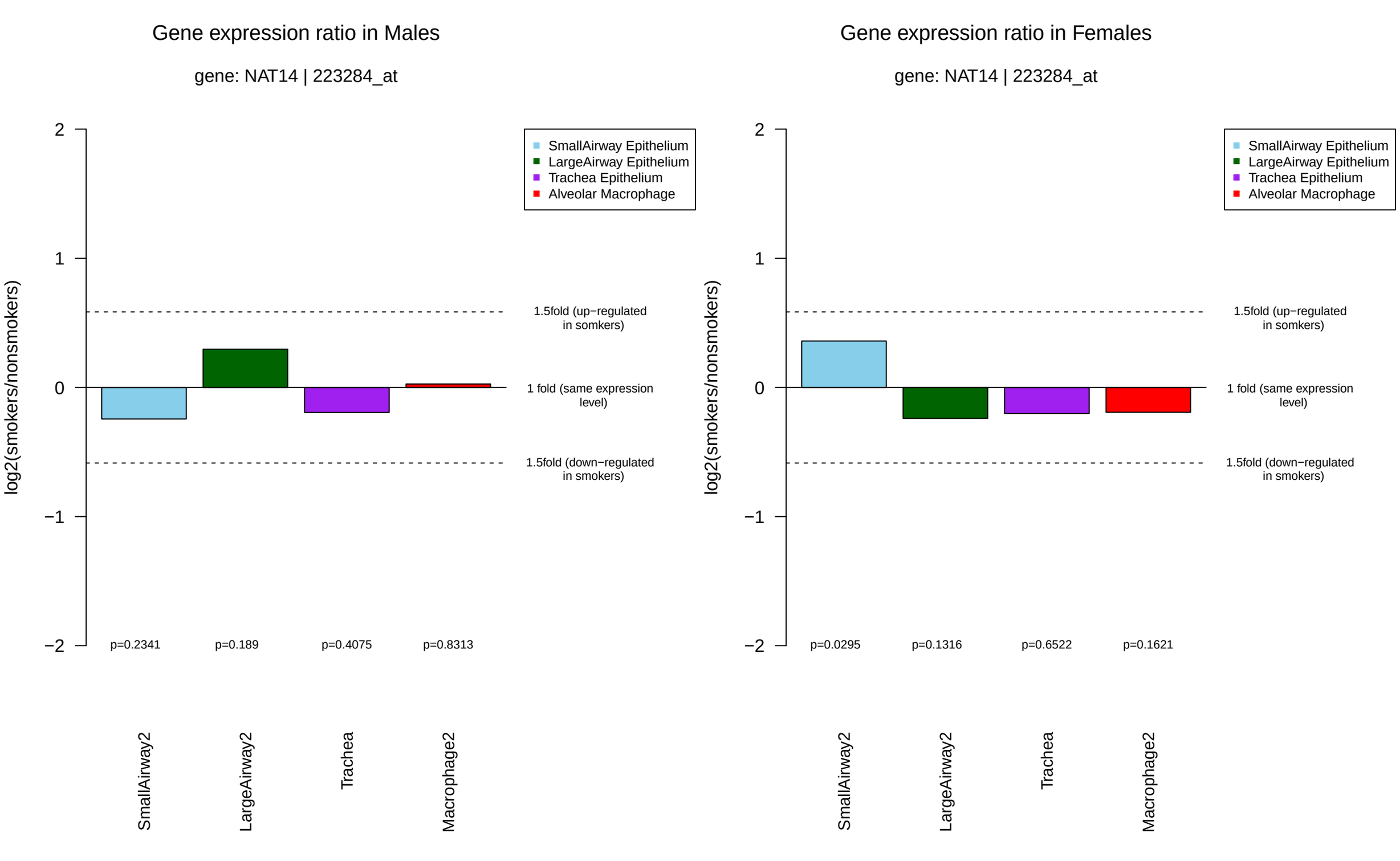
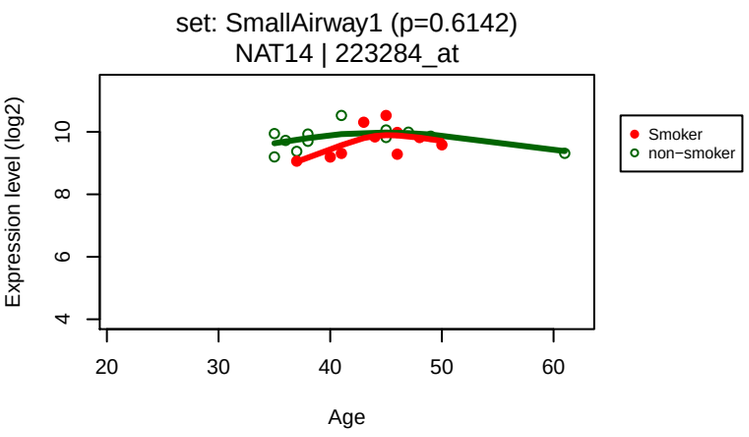
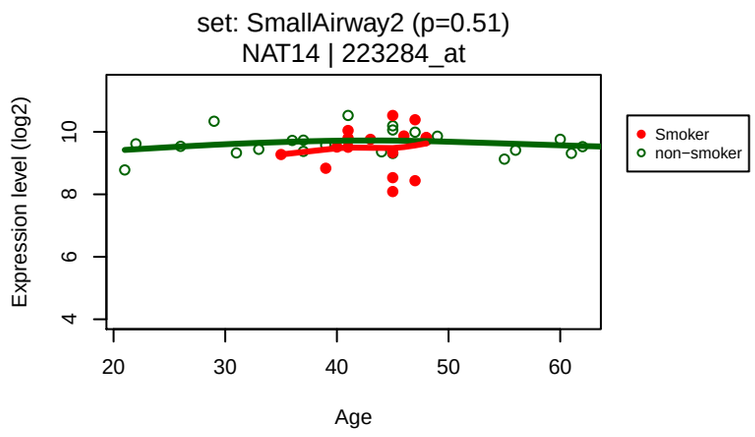
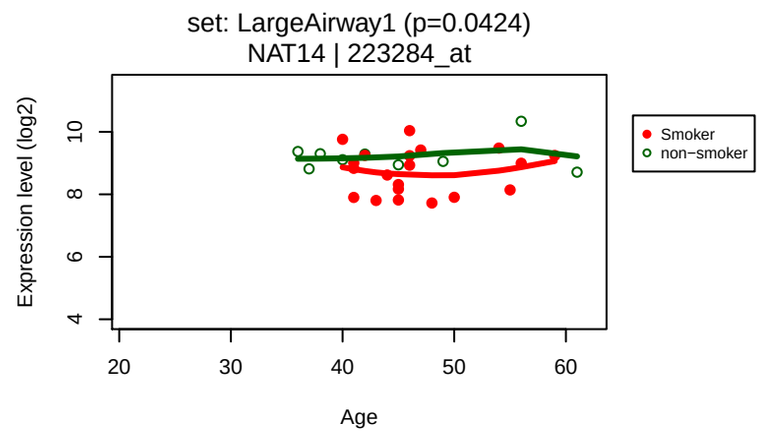

Gene expression of smokers and nonsmokers
gene: NAT14 | 223284_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6142
p=0.51
p=0.6354
p=0.0424
p=0.5756
p=0.2488
p=0.0715
p=0.7043
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

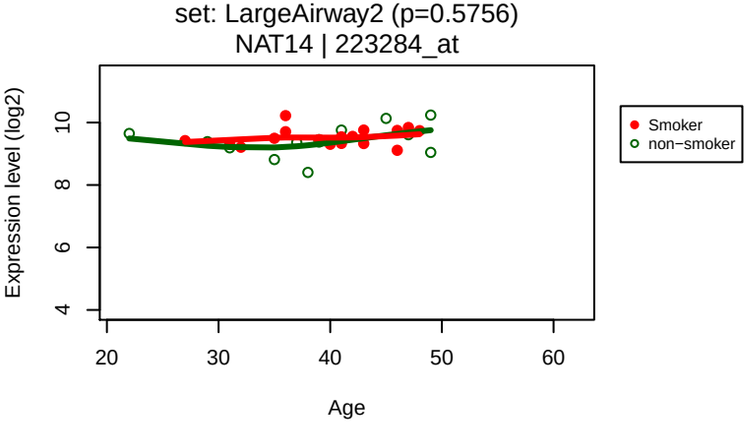

Gene expression ratio (smokers/nonsmokers)
gene: NAT14 | 223284_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6142
p=0.51
p=0.6354
p=0.0424
p=0.5756
p=0.2488
p=0.0715
p=0.7043
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

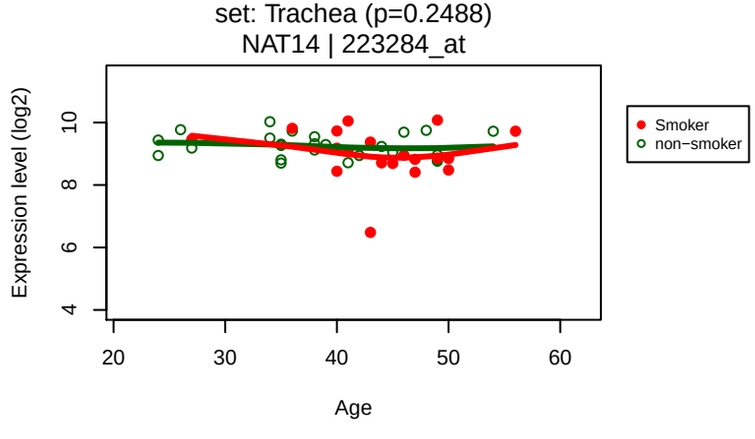

Gene expression ratio in Males
gene: NAT14 | 223284_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2341
p=0.189
p=0.4075
p=0.8313
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NAT14 | 223284_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0295
p=0.1316
p=0.6522
p=0.1621
SmallAirway2
LargeAirway2
Trachea
Macrophage2

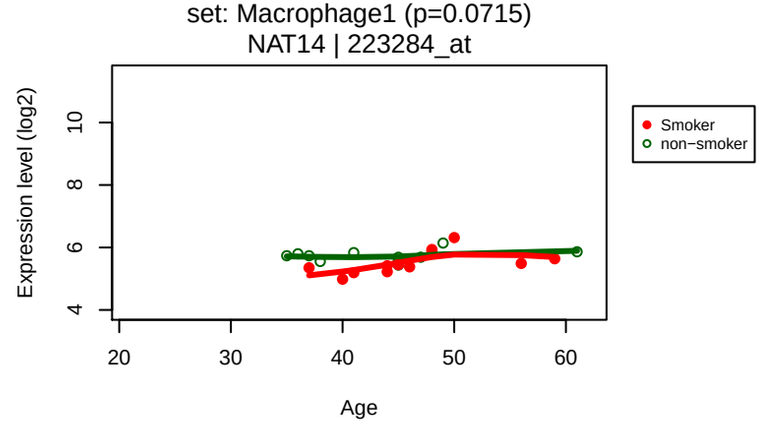

set: SmallAirway1 (p=0.6142)
NAT14 | 223284_at
Expression level (log2)
Age
Smoker
non−smoker

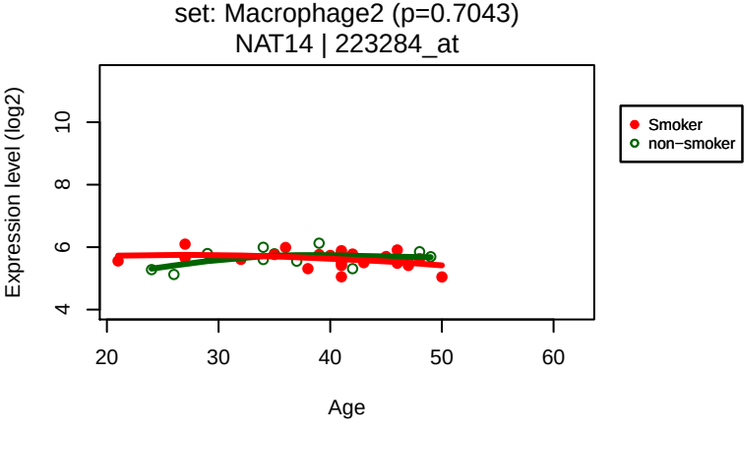

set: SmallAirway2 (p=0.51)
NAT14 | 223284_at
Expression level (log2)
Age
Smoker
non−smoker

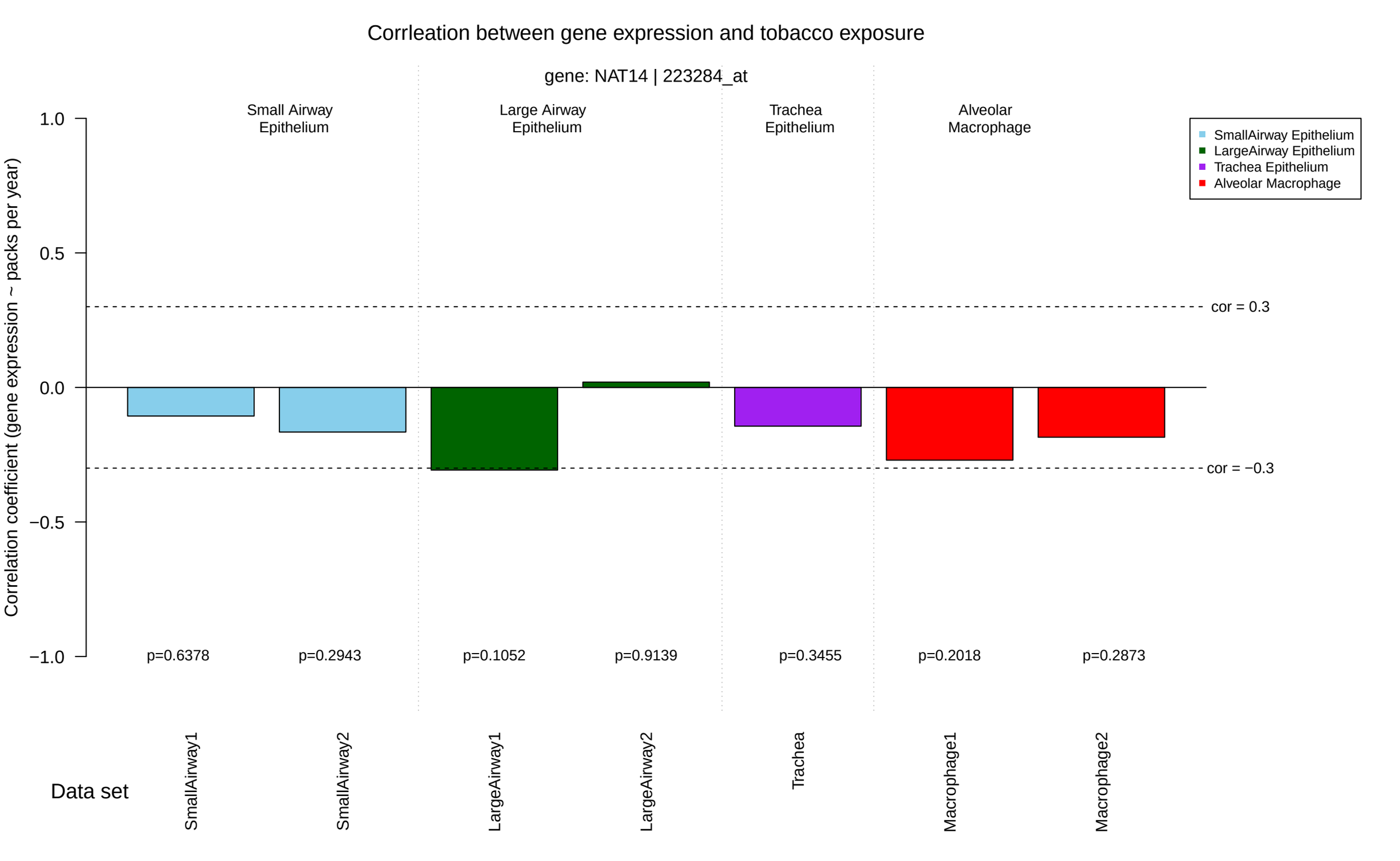

set: LargeAirway1 (p=0.0424)
NAT14 | 223284_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5756)
NAT14 | 223284_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2488)
NAT14 | 223284_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0715)
NAT14 | 223284_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7043)
NAT14 | 223284_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NAT14 | 223284_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6378
p=0.2943
p=0.1052
p=0.9139
p=0.3455
p=0.2018
p=0.2873
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2